RECEIVING STATION OH2GAX · NEAR HELSINKI-VANTAA

MODE-S Wind

Airliners as weather sensors

A passive 1090 MHz receiving station that turns the replies of passing airliners into **wind, temperature, windshear and turbulence** — measured by the aircraft themselves, live, from the final approach up to cruise level.

Passive · listen only | Live every few seconds | Web + phone | Open source · GPLv3



Otso Laakso · OH2GAX · Version v.01112026 · November 2026 · 1 / 2

≈ 130 /s — Mode S and ADS-B messages heard, 24-hour mean

160 NM — range of the receiver at cruise levels

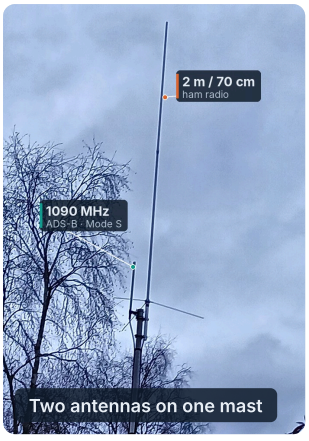
≈ 200 + 200 — landings and departures followed every day

0.5 s — a headwind value of each aircraft on final and climb-out

4.4 kt — wind against the MEPS model, vector RMS over 21 days

2 sites — the local station and the cloud server — the same software

1 HOW IT WORKS — one antenna, one receiver, one small computer; the users are served from the cloud







Aircraft 1 — Radars ask on 1030 MHz; the transponders answer on 1090 MHz with airspeed, heading, Mach, track and ground speed (Mode S EHS) and broadcast GNSS position and velocity (ADS-B).

Receiver 2 — Listens only: the Radarcape decodes every reply with a GPS-timed nanosecond stamp (MLAT).

Local station 3 — A Raspberry Pi 4 decodes, checks, calibrates and stores — also without the internet.

Cloud server 4 — A full second copy at AWS, fed by an outbound encrypted tunnel, serves the users over HTTPS.

You 5 — any browser and the phone view **/app** — the address on page 2

2 WHAT THE AIRCRAFT MEASURE — the same numbers air traffic control uses, turned into weather

Wind

```
wind = (GS, track) − (TAS, heading)
```

The ground-speed vector minus the air vector, from two replies at most 10 s apart; the magnetic heading turned to true with the World Magnetic Model.

Temperature

$$T = 288.15\ \mathrm{K} \cdot (\mathrm{TAS} / (a_0 \cdot M))^2$$

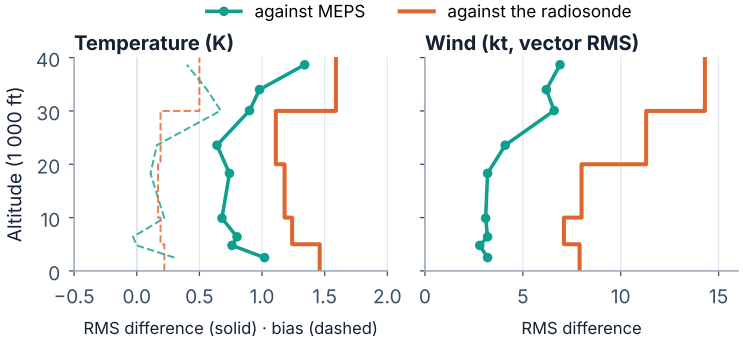
From the true airspeed and Mach number (a_0 = 661.47 kt); each reply is coarse, so profiles use trimmed means of many aircraft.

Heights of the pressure surfaces

$$z = \text{pressure altitude} + \Delta_{\mathrm{GNSS-baro}}$$

The pressure altitude says which surface the aircraft flies on, its ADS-B GNSS difference how high it lies: height contours and the geostrophic wind.

3 SELF-CORRECTING, INDEPENDENTLY CHECKED — against the radiosonde and the MEPS weather model



How well the aircraft agree — the system's own checks to 1 Nov 2026: the Mode S area profile against **MEPS** hourly (21 days, 298 hours) and the **Jokioinen radiosonde** (66 launches since 10 Oct); raw temperatures. The sonde flies ~100 km away: part of its difference is distance.

0.9 K · 4.4 kt — temperature · wind against MEPS (RMS, all levels)

1.3 K · 9.7 kt — against the radiosonde (RMS, 66 launches)

11.5 → 9.1 kt — **Heading offset of each airframe** — old magnetic-variation tables in the avionics, learnt per aircraft: wind against the soundings when introduced

1.27 → 0.96 K — **Upper-level temperature** — the fleet reads warm at 300–250 hPa; a correction learnt every hour: error against the soundings at 200–300 hPa

± 2 m — **GNSS heights** of the pressure surfaces against MEPS at 400–250 hPa, from altimeter and geoid corrections learnt at every sounding

© 2026 Otso Laakso (OH2GAX) · MODE-S Wind, GPLv3 · github.com/oh2gax/mode_s_wind

Decoding with pyModeS (Junzi Sun, TU Delft) · method after Siebren de Haan (KNMI)

4 LOW-LEVEL WINDSHEAR AND TURBULENCE — for every landing and departure at Helsinki-Vantaa (research)

What matters to an aircraft near the ground is a sudden change of the **headwind along its own path** — and that is exactly what each aircraft reports: its true airspeed minus its ground speed. The station follows it about **every 0.5 s** for every aircraft on final and on the climb-out (the airspeed from Mode S, the ground speed from ADS-B), finds headwind **ramps** as the Hong Kong Observatory does in its lidar scans, and compares each 200-ft band with the flights of the previous 45 minutes. Every flight gets a stored **windshear report**, and the turbulence (**EDR**, the ICAO measure) of its approach or climb-out from the same replies.

ADV advisory ≥ 10 kt **WSA** alert ≥ 15 kt **MBA** ≥ 30 kt loss

`15A WSA 15K+ 2MF`

"Runway one five arrival, windshear alert, one five knot gain, two mile final"

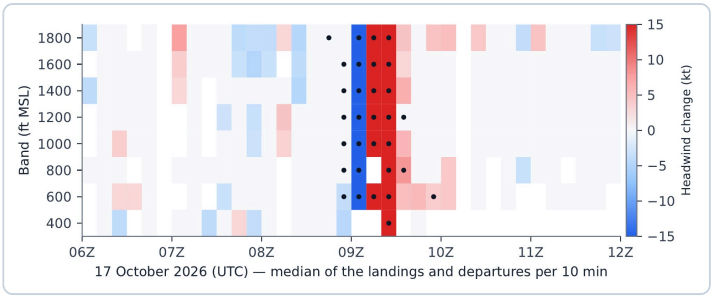
The boundary layer is followed too: the **low-level jet** and the **surface inversion** hour by hour, and every flight's own headwind profile from 200 to 5 000 ft as a column of a time–height section. The results are checked against the METAR windshear groups, the pilots' reports (ARS, WXREP) and the go-arounds — in a **study log** written for aviation meteorologists.

0.9 %
of the landings with an advisory, 9–27 Oct 2026

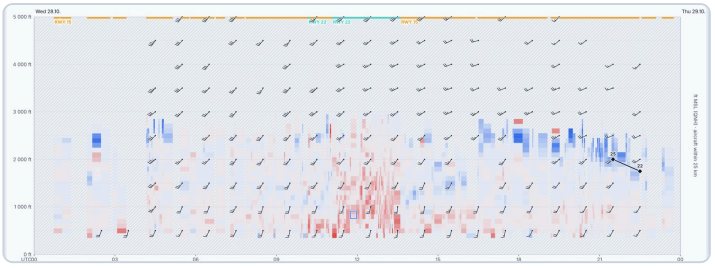
29 % · 2.4 %
of the landings in light · moderate turbulence (EDR), 9–29 Oct 2026

400–1 800 ft
200-ft bands compared flight to flight

16 %
of the night hours with a low-level jet, 29 Sep – 29 Oct 2026

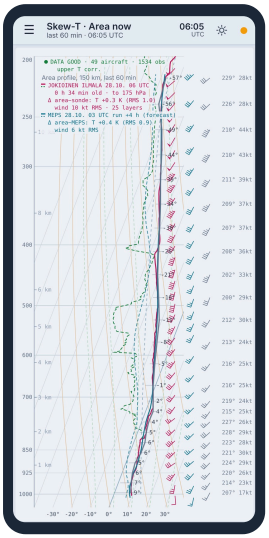


A front, band by band — 17 Oct 2026. The headwind change of the landings and departures against the previous 45 min: at 09 UTC a blue streak (less headwind) through every band from 600 to 1 800 ft, then the red reversal after the runway change. A dot: a flight at or beyond its advisory limit.



Wind shear flight by flight, 28 Oct 2026 (Sounding page): each landing a thin column, red where its headwind grows upwards, blue where it falls off — the friction layer by day, from 17 UTC the top of a low-level jet at 2 000–2 600 ft.

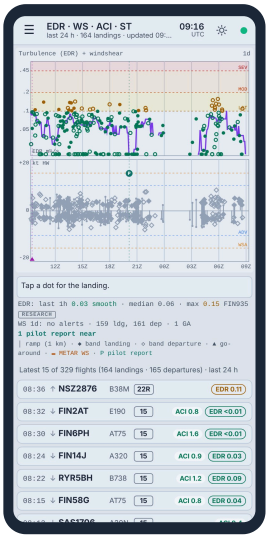
5 ON YOUR PHONE — mode-s-wind.oh2gax.org/app: one view at a time, no app store needed



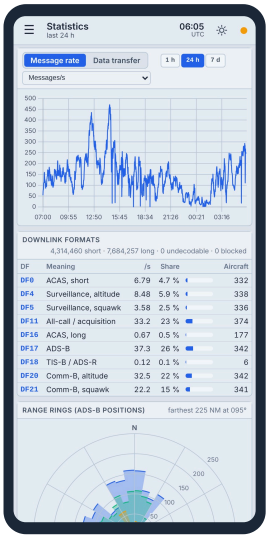
Skew-T · area now
49 aircraft of the last hour with the radiosonde and the model



ILS approach
the approach map with every flight's turbulence, the profile and strips



EDR · WS · ACI · ST
turbulence and windshear of 24 h, the pilot report (P), the flights



Statistics
what the receiver hears: rate, formats, range rings

mode-s-wind.oh2gax.org/app
scan with the phone's camera, then *Add to Home Screen*

- **Eight views** — Skew-T, time–height, ILS approach, turbulence and windshear, history, statistics, weather
- **Tap for values**, portrait or landscape, dark or light
- **Light on data** — about 1–4 MB an hour in the live ILS view

6 IN THE BROWSER — the full pages at mode-s-wind.oh2gax.org

Live Map
aircraft by meteo source, wind barbs, weather, receiver coverage; an ATC-style radar look

Sounding
Skew-T now or any hour, time–height up to 14 days, radiosonde and model checks

Wind Map
wind and temperature at a level, streamlines, heights, jet stream, cross-sections

Weather
MEPS maps 48 h ahead — rain, gusts, clouds, fog, 2 m temperature — and the FMI radar

Windshear
live ILS profile, alerts, go-arounds, approach history, EDR and boundary layer

GPS
area-wide GNSS jamming and spoofing seen from the aircraft

Statistics
formats, registers, interrogators, range, system health

See it live

https://mode-s-wind.oh2gax.org
phone: **/app** · access with a personal user name — ask the author

Open data FMI (soundings, radar) · MET Norway (MEPS) · NOAA (METAR) · WMM2025 · EGM96
Open source github.com/oh2gax/mode_s_wind (GPLv3) · user guides for every page

Research only. Passive and uncertified: the alerts describe air already flown through — not for flight operations or air traffic control.

© 2026 Otso Laakso (OH2GAX) · MODE-S Wind, GPLv3 · github.com/oh2gax/mode_s_wind